

JAST (Journal of Animal Science and Technology) TITLE PAGE

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ARTICLE INFORMATION	Fill in information in each box below
Article Type	Research article
Article Title (within 20 words without abbreviations)	Dietary crude protein level and arginine analogue supplementation affect growth performance, nutrient digestibility, ileal morphology, and serum metabolites in broilers
Running Title (within 10 words)	Arginine analogue in low crude protein broiler diets
Author	Myunghwan Yu ¹ , Hyunsoo Kim ¹ , Chunik Lim ¹ , Jun Seon Hong ¹ , Hee-Jin Kim ¹ , Yunseok Kim ¹ , Sung-June Byun ² , Eui-Chul Hong ^{1,*}
Affiliation	¹ Poultry Research Center, Department of Animal Resources Development, National Institute of Animal Science, Pyeongchang 25342, Korea ² Animal Genetic Resources Research Center, National Institute of Animal Science, Hamyang 50000, Korea
ORCID (for more information, please visit https://orcid.org)	Myunghwan Yu (https://orcid.org/0000-0003-4479-4677) Hyunsoo Kim (https://orcid.org/0000-0001-8887-1318) Chunik Lim (https://orcid.org/0000-0003-0386-5694) Jun Seon Hong (https://orcid.org/0000-0003-2142-9888) Hee-Jin Kim (https://orcid.org/0000-0002-6959-9790) Yunseok Kim (https://orcid.org/0000-0003-4186-7877) Sung-June Byun (https://orcid.org/0000-0001-6909-1025) Eui-Chul Hong (https://orcid.org/0000-0003-1982-2023)
Competing interests	No potential conflict of interest relevant to this article was reported.
Funding sources State funding sources (grants, funding sources, equipment, and supplies). Include name and number of grant if available.	This research was supported by the Cooperative Research Program for Agriculture Science and Technology Development (Project No. PJ01760002) of the Rural Development Administration, Republic of Korea.
Acknowledgements	This study was also supported by the 2026 RDA Fellowship Program of National Institute of Animal Science, Rural Development Administration, Republic of Korea. The arginine analogue used in the present study was supplied by the Precision Animal Nutrition Division of the National Institute of Animal Science (Wanju, Republic of Korea). The compound is associated with Korean patent application No. KR-10-2025-01936725, and has the CAS number 210545-23-6.
Availability of data and material	Upon reasonable request, the datasets of this study can be available from the corresponding author.
Authors' contributions Please specify the authors' role using this form.	Conceptualization: Yu M, Hong EC. Data curation: Yu M. Formal analysis: Yu M. Methodology: Yu M, Kim H, Lim C, Kim, HJ, Hong EC. Software: Yu M. Validation: Kim H, Kim Y, Byun SJ. Investigation: Yu M, Lim C, Hong JS, Kim HJ, Byun SJ, Hong EC. Writing - original draft: Yu M. Writing - review & editing: Yu M, Kim H, Lim C, Hong JS, Kim HJ, Kim Y, Byun SJ, Hong EC.
Ethics approval and consent to participate	This study was approved by the Institutional Animal Care and Use Committee (IACUC) of the National Institute of Animal Science, Rural Development Administration, Republic of Korea (Approval No.: NIAS 2025-0026).

CORRESPONDING AUTHOR CONTACT INFORMATION

For the corresponding author (responsible for correspondence, proofreading, and reprints)	Fill in information in each box below

First name, middle initial, last name	Eui-Chul Hong
Email address – this is where your proofs will be sent	drhong@korea.kr
Secondary Email address	-
Address	Poultry Research Center, Department of Animal Resources Development, National Institute of Animal Science, Rural Development Administration, Pyeongchang 25342, Republic of Korea
Cell phone number	None
Office phone number	+82-33-330-9555
Fax number	None

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Abstract

This study evaluated the effects of dietary crude protein (CP) level and arginine analogue supplementation on growth performance, nutrient utilization, intestinal morphology, and serum metabolites in broiler chickens during the finisher phase. A total of 256 male Ross 308 broilers were assigned to a 2×2 factorial design consisting of two dietary CP levels and two arginine analogue supplementation levels. From day 22 to day 35, birds were fed either a normal-CP diet containing 19% CP or a low-CP diet containing 17% CP, with or without arginine analogue supplementation. Each treatment included eight replicate cages with four birds per cage. Growth performance, apparent ileal digestibility (AID) of nutrients and amino acids, ileal morphology, and serum biochemical parameters were assessed on day 35. Broilers fed the normal-CP diet showed greater body weight on day 35, greater average daily gain, and a lower feed conversion ratio (FCR) during days 22–35 than those fed the low-CP diet ($p < 0.05$). Arginine analogue supplementation reduced FCR ($p < 0.05$) regardless of dietary CP level. Additionally, supplementation increased ($p < 0.01$) the AID of CP, arginine, and methionine, and improved ileal morphology ($p < 0.05$) by increasing villus height and crypt depth. Birds fed the low-CP diet exhibited higher ($p < 0.05$) AID values for several amino acids, including arginine, histidine, isoleucine, leucine, lysine, phenylalanine, valine, alanine, glycine, proline, and tyrosine. However, the low-CP diet also elevated ($p < 0.05$) serum alanine aminotransferase activity and decreased ($p < 0.05$) serum calcium concentration. Conversely, arginine analogue supplementation decreased ($p < 0.05$) serum triglyceride concentration. Overall, reducing dietary CP from 19% to 17% negatively affected growth performance, despite maintaining comparable levels of lysine, methionine plus cysteine, and threonine. However, arginine analogue supplementation improved feed efficiency, nutrient digestibility, and intestinal morphology. These findings suggest that arginine analogue supplementation may serve as an effective nutritional strategy to improve nutrient utilization and intestinal health in broilers fed reduced-CP diets, although additional amino acid optimization may be required to fully maintain growth performance.

Keywords: arginine analogue, broiler, feed efficiency, intestinal morphology, low crude protein diet, nutrient digestibility

Introduction

Feed constitutes the largest proportion of production costs in broiler operations, and protein and amino acids are among the most expensive nutritional components of broiler diets [1,2]. Consequently, increasing attention has been directed toward nutritional strategies that reduce the inclusion of costly protein ingredients without compromising productive performance [3]. Low-protein diets have been recognized as one such strategy because appropriate supplementation with crystalline amino acids can reduce excessive protein supply while maintaining growth performance [4,5]. Additionally, low-protein feeding is considered a practical approach for mitigating environmental burdens by reducing nitrogen excretion, thereby improving the economic and environmental sustainability of broiler production [6].

Arginine is an essential amino acid in poultry because birds lack a functional urea cycle and therefore cannot synthesize sufficient amounts endogenously [7]. Beyond its fundamental role in protein synthesis, arginine serves as a precursor for several biologically important metabolites, including nitric oxide, polyamines, creatine, citrulline, proline, and glutamate. Consequently, arginine is involved in growth, immune regulation, antioxidant defense, intestinal function, and metabolic homeostasis [8–10]. Previous studies in broilers have shown that increasing dietary arginine supply or the arginine-to-lysine ratio improves growth performance and feed efficiency [11,12]. Under low-protein conditions, arginine nutrition has also been associated with improvements in body weight, skin quality, and creatine-related variables [13]. Furthermore, arginine has been closely linked to intestinal barrier function, immune responses, oxidative stress regulation, and gut microbial homeostasis [14–16].

The nutritional importance of arginine may be particularly relevant in reduced-crude protein diets because decreasing the inclusion of protein-rich ingredients can reduce dietary arginine supply and alter the balance among amino acids. However, previous studies investigating arginine-related

nutritional strategies in broilers have primarily focused on conventional L-arginine or well-characterized arginine-related compounds, such as citrulline and guanidinoacetic acid [17–19]. Information regarding the application of other arginine-related compounds remains limited, particularly under reduced-crude protein conditions. The arginine analogue evaluated in the present study was selected as a novel candidate compound for the initial assessment of potential arginine-associated nutritional responses. However, its nutritional activity and physiological effects in broiler chickens have not been established. Therefore, the present study was conducted to evaluate the main and interactive effects of dietary crude protein level and arginine analogue supplementation on growth performance, nutrient utilization, intestinal morphology, and serum metabolites in finisher broiler chickens.

Materials and Methods

Experimental design and diets

A total of 256 one-day-old male Ross 308 broiler chickens were raised in battery cages (75 × 60 × 40 cm³), corresponding to a stocking density of 0.056 m² per bird, in accordance with the Ross 308 Broiler Management Guide until 21 days of age [20]. All chicks were fed a crumbled commercial pre-starter diet (ME: 3000 kcal/kg; CP: 22%) from day 1 to day 7, followed by a grower diet (ME: 3050 kcal/kg; CP: 20%) from day 8 to day 21. On day 22, a total of 128 birds with similar body weight (1007.93 ± 4.130 g) were selected from the initial population and randomly assigned to four treatment groups, each consisting of eight replicate cages with four birds per cage. The treatments were arranged in a 2 × 2 factorial design with two dietary crude protein (CP) levels (normal, 19%; low, 17%) and two arginine analogue supplementation levels (with or without supplementation). The normal-CP diet was formulated to meet or exceed the nutrient recommendations for broilers [21]. The arginine analogue was provided by the Precision Animal Nutrition Division, National Institute of Animal Science (Wanju, Republic of Korea). The ingredient composition, calculated nutrient values, and analyzed proximate composition of the finisher diets (days 22–35) are presented in Tables 1 and

2. Chromium oxide (Cr_2O_3 ; >99.95% purity, Sigma-Aldrich, St. Louis, MO, USA) was included at 0.3% in all diets as an indigestible marker for digestibility assays.

Bird management

Each cage was equipped with one nipple drinker and a plastic feed trough to provide feed and water efficiently. Birds were offered the experimental diets *ad libitum* and had unrestricted access to fresh drinking water throughout the study. Vaccinations against Newcastle disease and coccidiosis were administered on day 1. The ambient temperature was maintained at $30 \pm 1^\circ\text{C}$ from days 1 to 3, gradually reduced to $25 \pm 1^\circ\text{C}$ by day 14, and thereafter maintained at $24 \pm 1^\circ\text{C}$ until the end of the trial, according to the Ross 308 Broiler Management Guide [20]. Average relative humidity was maintained at $56 \pm 14\%$. A lighting schedule of 23 h light and 1 h darkness (23L:1D) was applied during the first week, followed by 18 h light and 6 h darkness (18L:6D) from day 7 to day 35.

Growth performance measurements

On day 35, body weight (BW) and residual feed in each cage were recorded to assess growth performance and feed consumption. Average daily gain (ADG) was determined by subtracting the initial and final BW measurements. Feed intake was obtained by calculating the difference between the amount of feed offered and the residual feed. Mortality was recorded daily throughout the experimental period. Mortality-adjusted average daily feed intake (ADFI) and feed conversion ratio (FCR) were computed on a cage basis.

Postmortem procedure and sample collection

On day 35, eight birds per treatment (one bird per cage) with body weights closest to the cage average were selected for blood and ileal tissue sampling. Blood samples were collected from the wing vein into serum separator tubes (SST; BD Biosciences, Franklin Lakes, NJ, USA). Serum was obtained by centrifugation at 3000 rpm and 4°C for 15 min and stored at -70°C until analysis of serum biochemical parameters. Following blood collection, birds were euthanized by exsanguination.

The abdominal cavity was opened, and the ileum was isolated from the digestive tract. The ileum was defined as the section extending from Meckel's diverticulum to the ileocecal junction [22]. A 3-cm segment of the ileum was excised, rinsed with phosphate-buffered saline (pH 7.4), and fixed in 10% formalin for histological analysis. For ileal digesta collection, two birds per cage with body weights closest to the cage mean were selected. Ileal digesta contents were collected into pre-labeled containers and stored at -70°C until nutrient digestibility analysis.

Nutrient digestibility

Digesta samples collected from the ileum were oven-dried at 55°C for 24 h, ground, and passed through a 0.75-mm sieve (ZM 200 Ultra-Centrifugal Mill, Retsch GmbH & Co. KG, Haan, Germany) prior to nutrient digestibility analysis, as described by [23]. Dry matter (DM) was determined according to AOAC (2005) official method 930.15. Chromium oxide concentrations were analyzed following the procedure described by [24]. Diets and digesta samples were analyzed for crude protein (method 984.13), crude fat (method 920.39), and amino acid (method 982.30 E [a,b]) as described by [25]. Gross energy contents of diets and digesta were determined using an adiabatic bomb calorimeter (IKA Calorimeter System C6000; IKA Works, Wilmington, NC, USA). The apparent ileal digestibility (AID) of DM, energy, CP, crude fat, and amino acids was calculated using the following equation:

$$\text{Digestibility, \%} = 100 - \left[100 \times \frac{M_{\text{diet}} \times N_{\text{digest}}}{M_{\text{digest}} \times N_{\text{diet}}} \right]$$

where M_{diet} represents the marker concentration in the diet; N_{digest} represents the nutrient concentration in the ileal digesta; M_{digest} represents the marker concentration in the ileal digesta, and N_{diet} represents the nutrient concentration in the diet.

Intestinal morphology

Fixed intestinal samples were embedded in paraffin, and sliced at $5\text{ }\mu\text{m}$ thickness, and stained with hematoxylin and eosin. The stained slides were examined using a fluorescence

microscope (TE2000, Nikon, Tokyo, Japan) equipped with a digital camera (DS-Ri2, Nikon, Japan). Morphological variables, including villus height, villus width, crypt depth, and the villus height-to-crypt depth ratio were measured via fluorescent images using NIS-Elements BR software (Nikon, Japan). Villus height was measured as the distance from the villus tip to its base, whereas villus width of the villi was measured at the midpoint of the villus height. Crypt depth was measured as the distance from the top of the crypt to the muscularis mucosa. Moreover, the villus absorptive surface area was calculated according to the method described by [26] using the following equation:

$$\text{Villus absorptive surface area} = 2\pi \times (\text{villus width} \div 2) \times \text{villus height}$$

Serum biochemistry

Serum biochemical parameters, including total cholesterol, triglycerides, glucose, total protein, aspartate aminotransferase (AST), alanine aminotransferase (ALT), creatinine, albumin, inorganic phosphorus (IP), lactate dehydrogenase (LDH), and calcium (Ca), were analyzed using an automatic biochemistry analyzer (AU480 Chemistry Analyzer, Beckman Coulter Inc., Brea, CA, USA).

Statistical analysis

Experimental data were analyzed using the general linear model (GLM) procedure for two-way ANOVA in SPSS software (version 28; IBM Corp., Armonk, NY, USA) to evaluate the main effects of dietary crude protein level and arginine analogue supplementation. The cage was considered the experimental unit for growth performance parameters, whereas individual birds served as the experimental unit for nutrient digestibility, intestinal morphology, and serum biochemical analyses. Statistical significance was declared at $p < 0.05$. When significant treatment effects were detected, means were separated using Tukey's multiple range test implemented in SPSS.

Results

Growth performance

The effects of dietary treatments on growth performance are presented in Table 3. Broilers fed the normal-CP diet had higher ($p < 0.05$) body weight at 35 days of age than those fed the low-CP diet. Similarly, birds receiving the normal-CP diet showed greater ($p < 0.05$) ADG during days 21–35 compared with those fed the low-CP diet. Broilers fed the low-CP diet exhibited a higher ($p < 0.001$) FCR than those fed the normal-CP diet during days 22–35. In addition, broilers receiving arginine analogue supplementation exhibited a lower ($p < 0.05$) FCR than those fed non-supplemented diets.

Nutrient digestibility

Table 4 illustrates the effects of dietary CP level and arginine analogue supplementation on the AID of nutrients in broiler chickens. On day 35, arginine analogue supplementation increased ($p < 0.01$) the AID of crude protein compared with the non-supplemented diets.

The results for the AID of amino acids are summarized in Tables 5 and 6. The AID of arginine, histidine, isoleucine, leucine, lysine, phenylalanine, valine, alanine, glycine, proline, and tyrosine was greater ($p < 0.05$) in birds fed the low-CP diet than in those fed the normal-CP diet. Furthermore, arginine analogue supplementation increased ($p < 0.01$) the AID of arginine and methionine compared with the non-supplemented diets. A significant interaction ($p < 0.05$) between dietary CP level and arginine analogue supplementation was observed for the AID of methionine.

Intestinal morphology

Table 7 shows the effects of dietary CP level and arginine analogue supplementation on ileal morphology in broiler chickens at day 35. Broilers receiving arginine analogue supplementation exhibited increased ($p < 0.05$) villus height and crypt depth than those fed non-supplemented diets.

Serum metabolites

Feeding the low-CP diet increased ($p < 0.01$) ALT activity compared with the normal-CP diet, whereas serum calcium concentration was higher ($p < 0.01$) in birds fed the normal-CP diet (Table 8). Furthermore, arginine analogue supplementation reduced ($p < 0.05$) serum triglyceride concentration compared with the non-supplemented diet.

Discussion

Reducing dietary CP levels through supplementation with crystalline amino acids has become a well-established strategy in broiler production for lowering feed costs and nitrogen excretion [3,27]. In the present finisher study, dietary CP was lowered from 19.0% to 17.0% (a 2.0 percentage-point reduction) on an isoenergetic basis (3,100 kcal/kg ME) using a single-stage corn–soybean meal formulation. The standardized ileal digestible (SID) levels of lysine (1.08%), threonine (0.72%), and methionine + cystine (0.86%) were equalized between diets through supplementation with L-lysine sulfate, DL-methionine, and L-threonine, whereas valine, isoleucine, arginine, and glycine were not supplemented in crystalline form. Under these formulation conditions, the analyzed arginine concentration declined from 1.29% to 1.15%, whereas the analyzed glycine + serine concentration decreased from 1.80% to 1.63%. Consequently, the analyzed total arginine-to-lysine ratio, calculated from the analyzed total amino acid composition presented in Table 2, fell from 0.98 to 0.88, and the glycine + serine pool decreased by approximately 9%. Accordingly, the present dietary model represented a moderate ideal-protein-based CP reduction in which lysine, sulfur amino acids, and threonine were maintained at adequate levels, while arginine and glycine + serine were not specifically supplemented. This formulation enabled evaluation of whether arginine analogue supplementation could compensate for the proportional decline of dietary arginine without modifying any other amino acid characteristics of the diet. Arginine is of particular nutritional importance in poultry because it is an indispensable amino acid and serves as a metabolic precursor for nitric oxide, polyamines, creatine, and proline [8]. However, most previous studies have focused on free L-arginine or closely related compounds such as guanidinoacetic acid [8,13], and information regarding

structurally modified arginine analogues in moderately reduced-CP corn–soybean meal diets remains limited. Therefore, the present 2 × 2 factorial experiment was designed to provide integrated evidence regarding growth performance, ileal nutrient digestibility, intestinal morphology, and serum biochemical responses under these dietary conditions, while also identifying research questions that warrant further investigation into the *in vivo* metabolic fate and physiological functions of the arginine analogue.

Birds fed the 19% CP diet had higher BW and ADG and lower FCR than those fed the 17% CP diet during days 22–35. This reduction in growth performance likely reflects changes in amino acid supply accompanying the 2-percentage-point reduction in dietary CP. Previous studies have similarly shown that reducing dietary CP can impair growth performance when amino acid supply is not adequately rebalanced [3,13,28]. In the present study, the basal diets were supplemented with crystalline lysine, methionine, and threonine, whereas the concentrations of several other amino acids declined as dietary CP was reduced. Notably, the analyzed total arginine concentration decreased from 1.29% in the 19% CP diet to 1.15% in the 17% CP diet, and the ratio of analyzed total arginine to analyzed total lysine (total Arg) in the 17% CP diet was 0.88. The analyzed glycine + serine concentration also decreased from 1.80% to 1.63%. Previous studies have reported that inadequate glycine-related amino acid supply in reduced-CP diets can adversely affect feed efficiency [29,30]. Therefore, the poorer growth performance observed in birds fed the 17% CP diet was likely associated with the combined reduction in amino acid supply rather than with CP reduction per se or limitation of a single amino acid. Against this nutritional background, arginine analogue supplementation reduced FCR irrespective of dietary CP level, although BW and ADG were not significantly affected. Previous studies using conventional L-arginine in reduced-CP diets have likewise reported improvements in feed efficiency with increasing arginine supply [13,31,32]. However, interpretation of these findings in relation to the present study requires caution because the total Arg ratio in our diets was calculated from the analyzed total concentrations of arginine and lysine, whereas the comparator studies expressed Arg ratios on an SID basis. Therefore, the total Arg ratio of 0.88 should be interpreted as an indicator of the relative proportion of analyzed total arginine

to analyzed total lysine within the present dietary formulation and should not be directly compared numerically with published SID Arg ratios. Accordingly, the similarity among studies lies in the association between increased arginine-related nutritional input and improved feed efficiency rather than in the absolute Arg ratios used. The absence of a significant CP $\times$ arginine analogue interaction further indicates that the improvement in FCR was not specific to the reduced-CP diet. Thus, although the FCR response observed in the present study is qualitatively consistent with previous observations following increased dietary arginine supply, the current data do not establish that the analogue is nutritionally equivalent to L-arginine or that its effect resulted from correction of an arginine deficiency. Direct determination of SID arginine and inclusion of an L-arginine reference treatment would be required to establish such relationships.

AID of CP increased with arginine analogue supplementation, whereas the 17% CP diet increased the AID of several indispensable and dispensable amino acids, including arginine, histidine, isoleucine, leucine, lysine, phenylalanine, valine, alanine, glycine, proline, and tyrosine. Similar increases in apparent amino acid digestibility have been reported in broilers fed reduced-CP diets [33,34], suggesting that the present response was not restricted to a specific amino acid. Because apparent digestibility is influenced by both dietary amino acid disappearance and endogenous amino acid flow, the higher AID observed with the 17% CP diet may partly reflect changes in the relative contribution of endogenous amino acids as dietary protein supply was reduced. In addition to its effect on CP digestibility, arginine analogue supplementation increased the AID of arginine and methionine. An increase in arginine digestibility following L-arginine supplementation has also been reported in broilers fed a reduced-CP diet [31], although direct comparison should be made cautiously because the present study evaluated an arginine analogue rather than conventional L-arginine. Of particular interest, methionine was the only amino acid showing a significant CP $\times$ arginine analogue interaction. Supplementation increased methionine AID by 4.02 percentage points in the 19% CP diet (90.04 to 94.06%) but by only 0.61 percentage points in the 17% CP diet (91.74 to 92.35%). This differential response indicates that the effect of the arginine analogue on methionine digestibility varied according to dietary CP level. However, because the interaction between the analogue and

methionine utilization was not directly examined, the nutritional basis for this response remains to be clarified.

Arginine analogue supplementation also increased ileal villus height and crypt depth, whereas dietary CP level had no significant effect on these morphometric variables. Previous studies have reported increased villus height following L-arginine supplementation in broilers fed reduced-CP diets [28,32], which is generally consistent with the villus response observed in the present study. However, the concurrent increase in crypt depth warrants cautious interpretation because crypt depth reflects changes in epithelial renewal and does not necessarily indicate improved intestinal function [33,34]. Moreover, the villus height-to-crypt depth ratio was not significantly affected by arginine analogue supplementation. Therefore, the present findings are more appropriately interpreted as morphological alterations in the ileal mucosa rather than as evidence of an overall improvement in intestinal health. Because epithelial turnover and barrier function were not directly assessed, the functional significance of these morphological changes remains uncertain.

Serum biochemical responses also differed according to dietary CP level and arginine analogue supplementation. The 17% CP diet increased serum ALT activity and decreased serum calcium concentration, whereas arginine analogue supplementation reduced serum triglyceride concentration. No significant CP $\times$ arginine analogue interactions were observed for these variables, indicating that the responses to dietary CP and analogue supplementation occurred independently within the present factorial design. Increased serum ALT activity has previously been reported in broilers fed reduced-CP diets [35], whereas different responses have been observed when additional amino acids were included in reduced-CP formulations [36]. These contrasting findings suggest that the ALT response may be influenced by the overall amino acid balance of the diet rather than by dietary CP concentration alone. The reduction in serum calcium observed with the 17% CP diet is difficult to interpret further because calcium retention, bone mineralization, and related indices of mineral utilization were not evaluated. In contrast, arginine analogue supplementation reduced serum triglyceride concentration without significantly affecting the other measured liver-related biochemical parameters. Previous studies with L-arginine have also reported changes in circulating lipid-related

indices in broilers [37]. Nevertheless, because hepatic lipid metabolism was not directly evaluated, the lower serum triglyceride concentration observed herein should be interpreted as a biochemical response to arginine analogue supplementation rather than as evidence of a specific lipid-metabolic mechanism. Future studies should establish the dose–response relationship of the arginine analogue and directly compare its nutritional effects with those of conventional L-arginine. Direct determination of SID arginine and methionine would help clarify the effects of the analogue on amino acid utilization, particularly the CP-dependent response observed for methionine digestibility. In addition, evaluation of intestinal barrier function and epithelial turnover, together with more comprehensive assessment of lipid and mineral metabolism, would help determine whether the morphological and serum biochemical changes observed in the present study translate into physiologically meaningful responses in broiler chickens.

Conclusion

In conclusion, reducing dietary CP from 19% to 17% during the finisher period negatively affected growth performance in broilers, despite maintaining comparable levels of SID lysine, methionine plus cysteine, and threonine. Supplementation with the arginine analogue improved feed conversion ratio, increased the apparent ileal digestibility of crude protein and selected amino acids, and enhanced ileal morphology, as indicated by increased villus height and crypt depth. In addition, arginine analogue supplementation reduced serum triglyceride concentration without causing adverse changes in the measured liver-related biochemical parameters. Collectively, these findings suggest that arginine analogue supplementation may represent a useful nutritional strategy for improving nutrient utilization and intestinal development in broilers fed reduced-CP diets. However, the analogue was unable to completely recover the growth performance reductions associated with the 17% CP diet, indicating that further optimization of dietary arginine and glycine + serine supply may be required in reduced-CP formulations. Future studies are needed to evaluate dose-dependent

responses of the arginine analogue and to investigate its influence on nitrogen excretion, standardized ileal amino acid digestibility, and metabolic pathways in broilers fed reduced-CP diets.

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469 Tables and Figures

470 **Table 1.** Ingredient and chemical composition of the experimental diets on an as-fed basis

Item	Normal diet	Low protein diet
Ingredient (%)		
Corn	66.89	73.13
Soybean meal (48%)	26.98	21.17
Soybean oil	2.76	1.93
Limestone	0.43	0.46
Di-calcium phosphate	1.40	1.44
L-lysine-Sulfate (70%)	0.32	0.50
DL-Methionine (98%)	0.32	0.37
L-Threonine (98%)	-	0.10
Salt	0.30	0.30
Vitamin-Mineral premix ¹	0.30	0.30
Cr ₂ O ₃	0.30	0.30
Calculated values		
Metabolizable energy	3,100	3,100
(kcal/kg)		
Crude protein	19.00	17.00
Calcium	0.65	0.65
Available phosphorus	0.36	0.36
SID lysine	1.08	1.08
SID methionine + cystine	0.86	0.86
SID methionine	0.58	0.61
SID threonine	0.72	0.72

471 ¹Provided per kilogram of diet: vitamin A (trans-retinyl acetate), 6,400 IU; vitamin D₃
472 (cholecalciferol), 1,760 IU; vitamin E, 12,000 IU; vitamin K₃, 1,760 mg; biotin, 80 mg; thiamin,
473 1,600 mg; riboflavin, 3,520 mg; pyridoxine, 2,400 mg; vitamin B₁₂, 8 mg; niacin, 28,000 mg;
474 pantothenic acid, 10,000 mg; folic acid 480 mg; Fe (from iron sulfate), 20,000 mg; Cu (from copper
475 sulfate), 81,000 mg; Zn (from zinc oxide), 32,500 mg; Mn (from manganese oxide), 43,800 mg; I
476 (from potassium iodide), 750 mg; Se (from sodium selenite), 50 mg.

477 SID, Standardized ileal digestible.

478

479 **Table 2.** Analyzed composition of the experimental diets

Item	Normal diet	Low protein diet
Dry matter (%)	88.56	88.78
Crude protein (%)	18.37	16.50
Gross energy (kcal/kg)	4018	3979
Crude fat (%)	5.20	4.50
Amino acids		
Lysine (%)	1.31	1.30
Methionine (%)	0.62	0.65
Cystine (%)	0.44	0.42
Threonine (%)	0.96	1.03
Arginine (%)	1.29	1.15
Histidine (%)	0.62	0.57
Isoleucine (%)	0.80	0.70
Leucine (%)	1.59	1.47
Phenylalanine (%)	0.95	0.84
Valine (%)	0.95	0.89
Alanine (%)	1.01	0.94
Aspartic acid (%)	1.87	1.63
Glutamic acid (%)	3.30	2.93
Glycine (%)	0.79	0.72
Proline (%)	1.16	1.10
Serine (%)	1.01	0.91
Tyrosine (%)	0.73	0.68

480

Table 3. Effects of dietary crude protein level and arginine analogue supplementation on growth performance in broilers¹.

Dietary treatment		BW (g)		ADG (g/day)	ADFI (g/day)	FCR (g/g)
CP level	Arginine analogue	Day 21	Day 35	Day 22-35	Day 22-35	Day 22-35
Normal	+	1007.34	2337.29	95.00	129.33	1.36
	-	1008.59	2310.78	93.01	127.39	1.38
Low	+	1008.44	2254.53	89.01	125.86	1.42
	-	1007.34	2215.26	86.28	134.75	1.56
SEM		4.130	19.873	1.437	1.650	0.017
Main effects						
CP level						
Normal		1007.97	2324.04	94.01	128.36	1.37
Low		1007.89	2234.90	87.64	130.31	1.49
SEM		3.991	19.441	1.406	1.699	0.019
Arginine analogue						
+		1007.89	2295.91	92.00	127.60	1.39
-		1007.97	2263.02	89.65	131.07	1.47
SEM		3.991	20.861	1.506	1.678	0.020
P-value						
CP level		0.993	0.033	0.035	0.560	<0.001
Arginine analogue		0.993	0.415	0.420	0.301	0.023
CP level × Arginine analogue		0.888	0.874	0.898	0.112	0.059

¹Values are presented as means of eight replicates per treatment.

CP, crude protein; BW, body weight; ADG, average daily gain; ADFI, average daily feed intake; FCR, feed conversion ratio; SEM, standard error of the mean.

Table 4. Effects of dietary crude protein level and arginine analogue supplementation on the apparent ileal digestibility of nutrients in broilers¹.

Dietary treatment		Nutrient digestibility (%)			
CP level	Arginine analogue	Energy	Crude protein	Crude fat	Dry matter
Normal	+	74.90	80.16	81.84	70.95
	-	74.15	76.45	82.43	71.85
Low	+	73.50	83.66	85.99	70.98
	-	71.02	77.54	81.00	69.10
SEM		0.717	0.758	0.738	0.869
Main effects					
CP level					
Normal		74.52	78.31	82.13	71.40
Low		72.26	80.60	83.50	70.04
SEM		0.712	0.866	0.783	0.850
Arginine analogue					
+		74.20	81.91	83.91	70.97
-		72.58	76.99	81.72	70.48
SEM		0.727	0.770	0.767	0.858
P-value					
CP level		0.126	0.141	0.364	0.441
Arginine analogue		0.270	0.003	0.148	0.779
CP level × Arginine analogue		0.552	0.433	0.069	0.432

¹Values are presented as means of eight replicates per treatment.
CP, crude protein; SEM, standard error of the mean.

Table 5. Effects of dietary crude protein level and arginine analogue supplementation on the apparent ileal digestibility of indispensable amino acids in broilers¹.

Dietary Treatment		Indispensable amino acids (%)								
CP level	Arginine analogue	Arginine	Histidine	Isoleucine	Leucine	Lysine	Methionine	Phenylalanine	Threonine	Valine
Normal	+	90.52	86.25	80.95	87.30	87.96	94.06 ^b	83.64	82.46	86.76
	-	88.01	85.53	81.14	86.64	87.42	90.04 ^a	82.88	81.29	87.66
Low	+	93.26	93.26	88.57	91.43	92.91	92.35 ^{ab}	93.04	82.24	94.20
	-	89.74	91.61	85.50	89.25	91.98	91.74 ^{ab}	91.03	80.76	92.93
SEM		0.521	0.406	0.718	0.493	0.384	0.319	0.581	0.574	0.471
Main effects										
CP level										
Normal		89.27	85.89	81.04	86.97	87.69	92.05	83.26	81.87	87.21
Low		91.50	92.43	87.04	90.34	92.45	92.05	92.03	81.50	93.57
SEM		0.575	0.409	0.722	0.498	0.378	0.405	0.578	0.567	0.466
Arginine analogue										
+		91.89	89.75	84.76	89.36	90.44	93.20	88.34	82.35	90.48
-		88.88	88.57	83.32	87.95	89.70	90.89	86.95	81.03	90.30
SEM		0.545	0.716	0.896	0.571	0.571	0.345	0.979	0.556	0.744
P-value										
CP level		0.041	<0.001	<0.001	0.002	<0.001	0.996	<0.001	0.747	<0.001
Arginine analogue		0.007	0.156	0.325	0.162	0.345	0.001	0.242	0.260	0.848
CP level × Arginine analogue		0.629	0.572	0.265	0.450	0.797	0.012	0.597	0.894	0.257

¹Values are presented as means of eight replicates per treatment.

CP, crude protein; SEM, standard error of the mean.

a,b Values within a row with different superscripts differ significantly ($p < 0.05$).

Table 6. Effects of dietary crude protein level and arginine analogue supplementation on the apparent ileal digestibility of dispensable amino acids in broilers¹.

Dietary Treatment		Dispensable amino acids (%)							
CP level	Arginine analogue	Alanine	Aspartic acid	Cystine	Glutamic acid	Glycine	Proline	Serine	Tyrosine
Normal	+	80.67	84.46	77.42	86.51	77.82	79.69	82.87	84.17
	-	81.01	84.30	78.38	84.32	78.44	80.60	83.33	84.04
Low	+	84.59	85.74	80.60	86.86	83.62	90.71	82.34	90.12
	-	83.00	85.78	79.23	84.83	81.67	89.00	80.24	87.95
SEM		0.616	0.693	0.693	0.652	0.707	0.508	0.592	0.59
Main effects									
CP level									
Normal		80.84	84.38	77.90	85.42	78.13	80.15	83.10	84.11
Low		83.79	85.76	79.92	85.85	82.65	89.85	81.29	89.03
SEM		0.605	0.67	0.678	0.659	0.696	0.506	0.588	0.587
Arginine analogue									
+		82.63	85.10	79.01	86.68	80.72	85.20	82.61	87.14
-		82.01	85.04	78.81	84.58	80.06	84.80	81.79	86.00
SEM		0.66	0.682	0.703	0.632	0.807	1.02	0.606	0.732
P-value									
CP level		0.023	0.327	0.157	0.744	0.003	<0.001	0.137	<0.001
Arginine analogue		0.621	0.968	0.884	0.117	0.644	0.696	0.495	0.339
CP level × Arginine analogue		0.440	0.942	0.407	0.953	0.372	0.206	0.291	0.394

¹Values are represented as means of eight replicates per treatment.
CP, crude protein; SEM, standard error of the mean.

Table 7. Effects of dietary crude protein level and arginine analogue supplementation on ileal morphology in broilers¹.

Dietary treatment		Ileal morphology				
CP level	Arginine analogue	Villus height (µm)	Crypt depth (µm)	Villus width (µm)	Villus height:Crypt depth ratio	Villus absorptive surface area (mm ²)
Normal	+	1032.42	137.86	133.63	7.54	0.43
Normal	-	963.28	124.07	128.50	7.81	0.39
Low	+	1038.55	136.00	121.85	7.73	0.40
Low	-	972.08	126.33	123.38	7.80	0.38
SEM		14.778	2.699	2.420	0.164	0.010
Main effects						
Crude Protein						
Normal		997.85	130.96	131.07	7.67	0.41
Low		1005.32	131.16	122.61	7.76	0.39
SEM		15.562	2.825	2.364	0.159	0.010
Arginine analogue						
+		1035.48	136.93	127.74	7.63	0.42
-		967.68	125.20	125.94	7.80	0.38
SEM		14.294	2.614	2.481	0.158	0.009
P-value						
CP level		0.802	0.970	0.092	0.790	0.243
Arginine analogue		0.029	0.038	0.712	0.605	0.107
CP level × Arginine analogue		0.964	0.706	0.497	0.762	0.566

¹Values are represented as means of eight replicates per treatment.

CP, crude protein; SEM, standard error of the mean.

Table 8. Effects of dietary crude protein level and arginine analogue supplementation on serum biochemical parameters in broilers¹.

Dietary treatment		Serum biochemical parameters										
CP level	Arginine analogue	TCHO (mg/dL)	TG (mg/dL)	GLU (mg/dL)	TP (g/dL)	AST (U/L)	ALT (U/L)	CREA (g/dL)	ALB (mg/dL)	IP (mg/dL)	LDH (mg/dL)	Ca (mg/dL)
Normal	+	135.54	69.96	290.40	2.98	266.95	1.51	0.26	1.27	8.65	2067.14	10.08
	-	153.20	75.79	276.39	2.82	257.70	1.35	0.25	1.23	8.44	1823.65	9.75
Low	+	148.38	65.43	267.76	2.98	266.30	1.96	0.26	1.19	8.25	2228.45	9.40
	-	150.81	89.26	289.13	2.73	255.40	1.78	0.25	1.19	8.51	2189.10	9.34
SEM		3.062	3.297	5.316	0.062	6.019	0.066	0.003	0.016	0.106	90.264	0.084
Main effects												
CP level												
Normal		144.37	72.88	283.39	2.90	262.33	1.43	0.26	1.25	8.55	1945.39	9.91
Low		149.59	77.34	278.44	2.86	260.85	1.87	0.26	1.19	8.38	2208.78	9.37
SEM		3.174	3.558	5.394	0.063	5.888	0.066	0.003	0.015	0.104	88.645	0.084
Arginine analogue												
+		141.96	67.69	279.08	2.98	266.63	1.74	0.26	1.23	8.45	2147.79	9.74
-		152.01	82.53	282.76	2.77	256.55	1.57	0.25	1.21	8.47	2006.38	9.54
SEM		3.076	3.315	5.402	0.06	5.817	0.075	0.003	0.016	0.105	90.936	0.096
P-value												
CP level		0.401	0.504	0.645	0.730	0.903	0.002	1.000	0.055	0.436	0.156	0.003
Arginine analogue		0.112	0.033	0.732	0.108	0.410	0.206	0.145	0.518	0.909	0.440	0.258
CP level × Arginine analogue		0.224	0.183	0.107	0.685	0.946	0.948	0.853	0.570	0.266	0.576	0.431

¹Values are presented as means of eight replicates per treatment.

CP, crude protein; TCHO, total cholesterol; TG, triglycerides; GLU, glucose; TP, total protein; AST, aspartate aminotransferase; ALT, alanine aminotransferase; CREA, creatinine; ALB, albumin; IP, inorganic phosphorus; LDH, lactate dehydrogenase; Ca, calcium; SEM, standard error of the mean.